

BIKASH ADHIKARI

Post-Doctoral Associate

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Summary

- **Plant science researcher** with over five years of experience improving crop resilience under abiotic stress, including heat, drought, and waterlogging, with strong expertise in source-sink relationships and nutrient uptake strategies to support fruit quality and yield.
- Hands-on research experience with a wide range of crops, including strawberry, leafy greens, peach, watermelon, corn, soybean, and cotton, across fields and controlled environments.
- Demonstrated excellence in scientific communication, undergraduate training, writing, and leadership through successful development of field and controlled environment trials, resulting in **20 peer-reviewed publications**, three articles across outreach and industry platforms, 50+ presentations, and multiple awards for research and communication.

Education

Ph.D.: Horticulture	2023
Mississippi State University, MS, USA	
M.S.: Horticulture	2020
University of Florida, FL, USA	
B.S.: Agriculture	2016
Tribhuvan University, Nepal	

Work Experience

Post-Doctoral Research Associate **08/2023-Present**

University of Florida (Dec 2024- present)

- Led multi-location organic strawberry and squash trials across Florida, assessing anaerobic soil disinfestation (ASD) and soil amendments to improve yield, flavor, and fruit/vegetable quality; research outcomes presented at the 2025 American Society for Horticultural Science.
- Investigated soil and plant health management strategies to reduce the soil-borne pathogens, weed pressure, and enhance vigor, firmness, and soluble solids in strawberries.
- Conducted multi-location field trials to assess the performance of grafted and non-grafted watermelon under varying nitrogen fertilizer rates and environmental conditions to optimize source-sink dynamics.

Mississippi State University (Aug 2023- Dec 2024)

- Led six innovative corn, soybean, and cotton stress physiology research projects in field, greenhouse, and controlled environments, resulting in one publication under revision and four under preparation; presented results at three conferences.
- Experience in pollination/pollen germination using a microscope to identify the variation in germination in pollens in corn subjected to different planting dates and nitrogen rates.
- Experience in designing and executing control environment, lab, field-scale scientific research, and analyzing and interpreting complex scientific data, as demonstrated by research on multi-crops (corn, soybean, and cotton) under waterlogging, chilling, high night temperature, heat, and drought stress at different growth stages and presented in three conferences.
- Effective communication and knowledge-sharing skills, as evidenced by 28 oral/poster presentations to scientific and general audiences and one award in a poster competition in the past year.

Bikash Adhikari | CV

- Demonstrated expertise in crop abiotic stress physiology, controlled environment agriculture, horticulture, agronomy, molecular genetics, remote sensing, and sustainability, as evidenced by 22 peer reviews for eight high-impact scientific journals.

Graduate Research Assistant

08/2018-08/2023

As a graduate student at Mississippi State University (Ph.D. candidate) and the University of Florida (master's degree)

- Designed and executed multidisciplinary field and controlled-environment trials on leafy greens, citrus, peach, cowpea, and other crops to evaluate physiological, biochemical, and molecular responses to abiotic stresses.
- Applied advanced statistical designs (20 graduate credits in experimental design) to M.S. and Ph.D. research, resulting in 14 peer-reviewed publications, 20+ conference presentations, and multiple national awards.
- Gained 5,000+ combined hours of greenhouse/CEA and field operations, including planting, crop care, pesticide/herbicide application, phenotyping, harvesting, and postharvest sample processing.
- Accumulated 3,500+ hours in laboratory research developing and applying protocols for antioxidant assays, ROS measurements, organic acids, and pigments (phenolics, anthocyanins, flavonoids).
- Conducted chlorophyll, carotenoid, and soluble sugar quantification via HPLC to screen waterlogging- and salt-tolerant genotypes of lettuce, cowpea, and kale.
- Operated agricultural machinery, automated irrigation/fertigation systems, and dataloggers for precision environmental monitoring and management.
- Actively engaged in ASHS and Tri-Society conferences, holding leadership roles, mentoring students, and disseminating research to scientific and grower audiences.

Micro-enterprise Development Officer

01/2017-05/2018

Karnali Integrated Rural Development and Research Center, Mugu, Nepal

- Conducted training on organic farming, vegetables, and several agronomic cropping systems for local growers to enhance their livelihood.
- Design and develop the cellar stores for storing apples produced in Mugu
- Other responsibilities include conducting field trials, collecting data, and organizing and facilitating dissemination activities through school visits, on-farm information sessions, and farm tours.
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Awards, Assistantships, and Travel Grants

- 3rd position for poster competition during the summer science and engineering symposium, Mississippi State University, 2024
- 1st position for poster competition during the summer science and engineering symposium, Mississippi State University, 2023
- 2nd position for poster competition during the spring graduate research symposium, Mississippi State University, 2023
- American Society for Horticultural Science National Travel Award (\$500), 2022
- American Society for Horticultural Science National Travel Award (\$500), 2021
- MAFES Graduate Research Assistantship Award (3 years: \$66,000), 2020
- American Society for Horticultural Science National Travel Award (\$500), 2019
- Graduate Student Council Travel Award (\$350), 2019
- Horticultural Sciences MS Graduate Research Assistantship (2 years: \$36,000), 2018

Memberships

- Member of Sigma Xi, the Scientific Research Honor Society (2025-2026)
- Member of the American Society for Horticultural Science (2018-present)
- Member of the Association of Horticulturists of the Indian Subcontinent (2018-present)
- Member of the American Society of Agronomy, the Crop Science Society of America, and the Soil Science Society of America (2023-2025)

Professional Experience

1. Coordinated a two-day LI-COR training workshop at the University of Florida, with technical support from the Department of Horticultural Sciences (Mar 2026)
2. Coordinated and presented the exhibit, "Adopting Grafting and Optimizing Fertilizer to Address Florida Watermelon Production Challenges," at the 2025 Suwannee Valley Watermelon Institute (Dec 2025)
3. Chair-elect for "History of horticulture science" and secretary for "Plant nutrient management" interest group at American Society for Horticultural Science (Aug 2025-Aug 2026)
4. Guest Editor for "Frontiers in Plant Science" and "Agronomy" peer-review journal (2025)
5. Participant in the "Preparing Future Faculty" for Fall 2025 at the University of Florida (Aug 2025-Dec 2025)
6. Volunteered as a judge for the five oral presentations at Controlled Environment Graduate Student Oral Competition, American Society for Horticultural Science (Aug 2025)
7. Obtained certificate for "Life Sciences Responsible Conduct of Research Course" required for NSF and NIFA grant conditions (Jul 2025)
8. Volunteered in multiple field days, including the Hydroponic Organic Tabletop Strawberry Research Field Day and Watermelon Field Day, where I shared research findings with growers, extension personnel, and industry stakeholders (Feb and May 2025)
9. Volunteered as a judge for the six poster and eight oral presentations at the Southern Branch of the American Society of Agronomy (Feb 2024)
10. Participated in 1-day LICOR-6800 Remote Photosynthesis training (Jan 2024)
11. Grant Writers' seminars and workshops focused on proposal writing, organized by Research and Economic Development at Mississippi State University (Sep 19, 2023)
12. Participated in the 3-day "Greenhouse Crop Production and Engineering Design Short Course" organized by the University of Arizona's Controlled Environment Agriculture Center (March 2023)
13. Participated in the short course: CEA Hydroponic Systems by Hort Americas (March 2023)
14. Participated in the short course: Hydroponic Strawberry Production by Hort Americas (March 2023)
15. One of the keynote speakers on the '1st Annual Greenhouse Vegetable Short Course' at North Mississippi Research & Extension Center in Verona, MS, about the management practices of lettuce in the controlled environment agriculture (March 2023)
16. Volunteered as a moderator for oral presentation and judge for the poster presentation at the Mississippi Academy of Sciences conference (2023)
17. LICOR-6800 Remote Photosynthesis training (Sep 2020)
18. Enrolled in an Agriculture Remote sensing course offered at Mississippi State University (Aug- Dec 2021)
19. Attended Training on Gene Expression Analysis and RNA extraction by Dr. Lisa Tang, Florida (2018)
20. Conducted training on Self-help group orientation and logistic support given to Dalit community members, Good Neighbors International, Nepal (2017)
21. Conducted four days of training on Cooperative and Bookkeeping for Good Neighbors International, Nepal (2017)
22. Conducted Ginger and turmeric farming and vegetable training to the farmers of 2 VDSCs of Mugu, Nepal (2017)

23. Conducted an Interaction workshop with marketing actors, Good Neighbors International, Nepal (2017)

Certifications

Category X: Pesticide Applicator Certification for the demonstration and research of pest control (oversee weed, pest, and disease scouting and management)

Grant proposal preparation experience

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- Grant Writers' seminars and workshops focused on proposal writing, organized by Research and Economic Development at Mississippi State University (Sep 19, 2023)
 - Harnessing Genetic Potential of Quinoa to Optimize Yield and Nutrition for Challenging Environments, 2024 (2025 MAFES SRI proposal: Not granted)
 - Navigating Floods: Unraveling the interplay of cultivar selection, crop rotation, cover crops, and nitrogen (2025 MAFES SRI proposal: Not granted)

Research Contributions

Publications: 20 peer-reviewed journal articles, one extension article, two international symposium article, and one industry magazine article.

1. **Adhikari, B.**, Sankarapillai, L.V., Dhillon, J.S., Reddy, K.N., Reddy, K.R., & Bheemanahalli, R. (2026). Quantifying resilience of major field crops to waterlogging during vegetative and reproductive stages. *Agricultural Water Management*.
2. **Adhikari, B.**, Olorunwa, O. J., Wilson, J., Barickman, T. C., Walters, K.J., Samiappan, S.K., & Bheemanahalli, R. (2025). Morpho-physiological, leaf hyperspectral, and nutritional responses of lettuce to salt stress. *Plant Physiology Reports*
3. Sankarapillai, L.V., **Adhikari, B.**, Silva, C.A., Wijewardane, N.K., & Bheemanahalli, R. (2025). Recovery response of soybean to waterlogging during the flowering and early-seed filling stages. *Discover Agriculture*
4. Poudel, S., Sankarapillai, L.V., **Adhikari, B.**, Reddy, K.R., & Bheemanahalli, R. (2025). Tolerance of cowpea (*Vigna unguiculata* (L.) Walp.) genotypes to drought stress during vegetative and reproductive stages. *Journal of Agronomy and Crop Science*
5. Sankarapillai, L.V., **Adhikari, B.**, Bista, M.K., Shrestha, A., Stetina, S.R., Reddy, K.R., & Bheemanahalli, R. (2025). High night temperature disrupts the assimilate utilization and yield potential in soybean. *Plant Stress*
6. Bista, M.K., **Adhikari, B.**, Sankarapillai, L.V., Peralisi, B., Reddy, K.R., Jenkins, J., & Bheemanahalli, R. (2024). Drought and heat stress induce differential physiological and agronomic trait responses in cotton. *Industrial Crop and Products*
7. **Adhikari, B.**, Olorunwa, O. J., Horgan, T. E., Wilson, J., Barickman, T. C., Li, T., & Bheemanahalli, R. (2024). Seed priming attenuates the impact of salt stress and enhances lettuce yields. *Journal of Agriculture and Food Research*
8. **Adhikari, B.**, Olorunwa, O.J., Wilson, J., & Barickman, T. C. (2024). Seasonal dynamics of lettuce growth on different electrical conductivity under the nutrient film technique hydroponic system. *Technology in Horticulture*
9. **Adhikari, B.**, Olorunwa, O. J., Brazel, S., Barickman, T. C., & Bheemanahalli, R. (2023). Impact of salt stress on physiology, leaf mass, and nutrient accumulation in romaine lettuce. *Photosynthetica*
10. Poudel, S., **Adhikari, B.**, Dhillon, J., Reddy, K. R., Stetina, S. R., & Bheemanahalli, R. (2023). Quantifying the physiological, yield, and quality plasticity of Southern USA soybeans under heat stress. *Plant Stress*
11. Olorunwa, O. J., **Adhikari, B.**, Brazel, S., Bheemanahalli, R., Barickman, T. C., & Reddy, K. R. (2023). Waterlogging stress reduces cowpea *Vigna unguiculata* L.) genotypes' growth, seed yield,

- and quality at different growth stages: Implications for developing tolerant cultivars under field conditions. *Agricultural Water Management*.
12. S.R. Brazel, **Adhikari B.**, and T.C. Barickman (2025). Temperature and short-term waterlogging impact the degradation of plant pigments in kale (*Brassica oleracea* L.) extracted samples. *ActaHort*.
 13. Olorunwa, O.J., **Adhikari, B.**, Brazel, S., Popescu, S., Popescu, G., and Barickman, T.C., 2022. Short Waterlogging Events Differently Affect Morphology and Photosynthesis of Two Cucumber (*Cucumis sativus* L.) Cultivars. *Frontiers in Plant Science*.
 14. Olorunwa, O. J., **Adhikari, B.**, Brazel, S., Popescu, S. C., Popescu, G. V., Shi, A., & Barickman, T. C. 2022. Waterlogging during the reproductive growth stage causes physiological and biochemical modifications in the leaves of cowpea (*Vigna unguiculata* L.) genotypes with contrasting tolerance. *Plant Physiology and Biochemistry*.
 15. **Adhikari, B.**, Olorunwa, O.J., and Barickman, T.C., 2022. Seed priming enhances seed germination and morphological traits of *Lactuca sativa* L. under Salt Stress. *Seeds*
 16. Olorunwa, O.J., **Adhikari, B.**, Brazel, S., Shi, A., Popescu, S.C., Popescu, G.V. and Barickman, T.C., 2022. Growth and Photosynthetic Responses of Cowpea Genotypes under Waterlogging at the Reproductive Stage. *Plants*
 17. **Adhikari, B.** and Vashisth, T., 2022, February. 'Just the right amount' of hydrogen cyanamide for profitable bud-break in low-chill peach cultivars. In XIV International Symposium on Plant Bioregulators in Fruit Production
 18. Barickman, T.C., **Adhikari, B.**, Sehgal, A., Walne, C.H., Reddy, K.R. and Gao, W., 2021. Drought and elevated carbon dioxide impact the morphophysiological profile of basil (*Ocimum basilicum* L.). *Crops*
 19. Barickman, T.C., **Adhikari, B.**, Sehgal, A., Walne, C.H., Reddy, K.R. and Gao, W., 2021. Drought and elevated CO₂ impact photosynthesis and biochemicals of basil (*Ocimum basilicum* L.). *Stresses*
 20. Olorunwa, O.J., **Adhikari, B.**, Shi, A. and Barickman, T.C., 2022. Screening of cowpea (*Vigna unguiculata* (L.) Walp.) genotypes for waterlogging tolerance using morpho-physiological traits at the early growth stage. *Plant Science*
 21. **Adhikari, B.**, Olorunwa, O.J., J. Wilson, and T.C. Barickman. 2021. Morphological and Physiological Response of Different Lettuce Genotypes to Salt Stress. *Stresses* 4: 285-304. Special Issue: Stress response in crops.
 22. **Adhikari, B.**, Dutt, M., & Vashisth, T. (2021). Comparative phytochemical analysis of the fruits of four Florida-grown finger lime (*Citrus australasica*) selections. *LWT*
 23. Archer, L., Brewer, M., **Adhikari, B.**, Esteves, E., Vincent, C., and Vashisth, T., 2021. Management Options for Improving Flowering in Citrus Production: HS1399, 12/2020. *EDIS*, 2021
 24. Vashisth T. and **Adhikari B.** Can a Lower Rate of Hydrogen Cyanamide Be Used in Peaches? *VSCNews. Specialty Crop Industry*. <https://specialtycropindustry.com/can-a-lower-rate-of-hydrogen-cyanamide-be-used-in-peaches/>

Under review and preparation (7):

1. Xu, N., **Adhikari B.**, Vincent, I., Moreira, D., Desaegeer, J., Özbudak, E., Hong, J., Roskopf, E.N., & Zhao, X. (2026). Assessing Anaerobic Soil Disinfestation for Organic Strawberry Production. *Scientia Horticulturae*. (Under preparation)
2. **Adhikari B.**, Vincent, I., Moreira, D., Xu, N., Desaegeer, J., Özbudak, E., Hong, J., Roskopf, E.N., & Zhao, X. (2026). Assessing Anaerobic Soil Disinfestation for Organic Strawberry Production. *Scientia Horticulturae*. (Under preparation)
3. Xu, N., Vincent, I., Moreira, D., Xu, N., Desaegeer, J., Özbudak, E., Hong, J., Roskopf, E.N., **Adhikari, B.**, & Zhao, X. (2026). Advances in Anaerobic Soil Disinfestation Research and Future Prospects. *The Science Total Environment*. (Under preparation)

4. **Adhikari, B.**, Vincent, I., Moreira, D., Xu, N., Desaeger, J., Hong, J., Roskopf, E.N., & Zhao, X. (2026). Root to Shoot: Strawberry Cultivar Growth Responses to Anaerobic Soil Disinfestation. *Journal of Horticultural Science*. (Under preparation)
5. **Adhikari, B.**, & Bheemanahalli, R. (2025). Chilling Tolerance of Corn Genotypes During Germination and Early Vegetative Stage. *Crop Design*. (Submitted)
6. **Adhikari, B.**, Sharma, R.K., Dhillon, J.S., & Bheemanahalli, R. (2025). Optimizing Nitrogen Fertilization for Enhanced Reproductive Success and Yield in Late-Planted Corn. *Field Crops Research*. (Under preparation)
7. Poudel, S., **Adhikari, B.**, Dhillon, J., Reddy, K. R., & Bheemanahalli, R. (2025). Complex response mechanism of soybean to interactive heat and drought stresses. *Plant Stress*. (Submitted).

Submitted abstracts (58):

1. **Adhikari, B.**, Vincent, I.R., Sanchez-Jones, T., Hochmuth, R., Warren, M., & Zhao, X. (2026). Examining Grafted Watermelon Production in Response to Different Nitrogen Fertilization Programs. *American Society of Horticultural Sciences Annual*. Aug 3-7, 2026 Dallas, TX (Oral presentation)
2. **Adhikari, B.**, Xu, N., Vincent, I.R., Moreira, D., Desaeger, J.A., Özbudak, E., Hong, J.C., Roskopf, E.N., & Zhao, X. (2026). Exploring the Legacy Impact of Anaerobic Soil Disinfestation Following Organic Strawberry Production. *American Society of Horticultural Sciences Annual*. Aug 3-7, 2026 Dallas, TX (Oral presentation)
3. Özbudak, E., Freitas, T., Olmedo, G., Bai, J., **Adhikari, B.**, Vincent, I.R., Moreira, D., Xu, N., Zhao, X., Cano, L.M., Hong, Roskopf, E.N., & J.C. (2026). Multi-Omics Characterization Of ASD-Treated Strawberry Links Molecular Changes To Enhanced Yield And Stress Responses. *APS Plant Health*. Aug 1-4, 2026, Providence, RI
4. Yifan, J., **Adhikari, B.**, and Zhao, X. (2026). Assessing water conservation potential of a tabletop strawberry production system in high tunnels. *10th UF Water Institute Symposium*. Feb 24-26, 2026.
5. **Adhikari, B.**, Vincent, I.R., Moreira, D., Xu, N., Desaeger, J.A., Özbudak, E., Hong, J.C., Roskopf, E.N., & Zhao, X. (2026). Application of Anaerobic soil disinfestation in organic strawberry production.
6. Zhao, X., **Adhikari, B.**, Vincent, I.R., & Sanchez-Jones, T. (2026). Cost-effective use of grafting to improve productivity and nitrogen-use efficiency of watermelon. *International Society for Horticultural Sciences*.
7. **Adhikari, B.**, Vincent, I.R., Moreira, D., Xu, N., Desaeger, J.A., Özbudak, E., Hong, J.C., Roskopf, E.N., & Zhao, X. (2025). Assessing Anaerobic Soil Disinfestation for Organic Strawberry Production. *Annual International Research Conference on Methyl Bromide Alternatives and Emissions Reduction*.
8. Özbudak, E., **Adhikari, B.**, Vincent, I.R., Moreira, D., Xu, N., Zhao, X., Cano, L., Roskopf, E.N., & Hong, J.C. (2025). Transcriptomic analysis of ASD-treated Florida strawberry. *Annual International Research Conference on Methyl Bromide Alternatives and Emissions Reduction*.
9. **Adhikari B.**, Vincent, I.R., Moreira, D., Xu, N., Desaeger, J.A., Hong, J.C., Roskopf, E.N., & Zhao, X. (2025). Anaerobic Soil Disinfestation with Organic Amendments Improves Strawberry Yield and Quality. *American Society of Horticultural Sciences Annual Conference (Oral Competition)*
10. **Adhikari, B.**, Vincent, I.R., Moreira, D., Xu, N., Desaeger, J.A., Hong, J.C., Roskopf, E.N., & Zhao, X. (2025). Root to Shoot: Strawberry Cultivar Growth Responses to Anaerobic Soil Disinfestation. *American Society of Horticultural Sciences Annual Conference*. (Poster Competition)
11. **Adhikari B.**, Sankarapillai, L. V., Dhillon J.S., Reddy, K.N., Reddy, K.R., & Bheemanahalli, R. (2024) Row crops response to waterlogging during vegetative and reproductive stages. *ASA, CSSA, SSSA International Annual Meeting*, San Antonio, Texas (Oral presentation).
12. Bista, M. K., **Adhikari B.**, Sankarapillai, L. V., Pieralisi, B., Reddy, K. R., Jenkins, J., & Bheemanahalli, R. (2024) Effects of drought, heat and their interaction during flowering and boll

- development in cotton. ASA, CSSA, SSSA International Annual Meeting, San Antonio, Texas (Oral presentation).
13. Sankarapillai, L. V., **Adhikari B.**, Stetina, S. R., Reddy, K. R., & Bheemanahalli, R. (2024). Does warmer nighttime affect the carbon balance in soybeans? ASA, CSSA, SSSA International Annual Meeting, San Antonio, Texas (Oral presentation).
 14. **Adhikari B.**, & Bheemanahalli, R. (2024) Chilling tolerance during germination and early vegetative stage in corn. ASA, CSSA, SSSA International Annual Meeting, San Antonio, Texas (Poster presentation).
 15. Sankarapillai, L. V., **Adhikari B.**, & Bheemanahalli, R. (2024). Does soybean tolerant or resilient to waterlogging stress? ASA, CSSA, SSSA International Annual Meeting, San Antonio, Texas (Poster presentation).
 16. **Adhikari B.**, Sankarapillai, L. V., Dhillon J.S., Reddy, K.N., Reddy, K.R., & Bheemanahalli, R. (2024) Row crops response to waterlogging during vegetative and reproductive stages, 6th MAS Summer Research Symposium, Mississippi State, MS. (Poster, III place).
 17. Hosahalli, V., **Adhikari B.**, Poudel, S., Reddy, K. R. & Bheemanahalli, R. (2024). Chilling tolerance of diverse soybean genotypes during seedling emergence, 6th MAS Summer Science and Engineering Symposium, Mississippi State, MS. (Poster presentation).
 18. Sankarapillai, L. V., **Adhikari B.**, Stetina, S. R., Reddy, K. R., & Bheemanahalli, R. (2024). Warmer nighttime disrupts the day-night assimilate utilization and yield potential in soybean. 6th MAS Summer Research Symposium, Mississippi State, MS. (Poster, I place).
 19. **Adhikari B.**, Sankarapillai, L. V., Dhillon J.S., Reddy, K.N., Reddy, K.R., & Bheemanahalli, R. (2024). Soybean response to early- and late-season waterlogging. 88th MAS Annual Meeting, Hattiesburg, MS. (Oral presentation).
 20. Bista, M. K., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024). Drought resilience of cotton cultivars during the reproductive stage. 88th MAS Annual Meeting, Hattiesburg, MS. (Oral, III Place)
 21. Poudel, S., Sankarapillai, L. V., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Effects of drought stress on physiology, growth and development of cowpea 88th MAS Annual Meeting, Hattiesburg, MS. (Oral presentation).
 22. Sankarapillai, L. V., **Adhikari B.**, Stetina, S. R., Reddy, K. R., & Bheemanahalli, R. (2024). Impact of high night temperature on soybean physiology and yield parameters. 88th MAS Annual Meeting, Hattiesburg, MS. (Oral presentation).
 23. Bista, M. K., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Effects of heat stress on cotton physiology and yield. 88th MAS Annual Meeting, Hattiesburg, MS. (Poster presentation).
 24. Poudel, S., Sankarapillai, L. V., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Effects of drought stress on physiology, growth and development of cowpea 88th MAS Annual Meeting, Hattiesburg, MS. (Poster presentation).
 25. Bista, M. K., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Quantifying the impacts of drought stress on cotton cultivars during the reproductive stage. Spring Graduate Research Symposium, Mississippi State University, Mississippi State, MS. (Oral presentation).
 26. Poudel, S., Sankarapillai, L. V., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Morpho-physiological and yield responses of cowpea to drought stress. Spring Graduate Research Symposium, Mississippi State University, Mississippi State, MS. (Oral presentation).
 27. Bista, M. K., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Quantifying the impacts of heat stress on cotton physiology and yield. Spring Graduate Research Symposium, Mississippi State University, Mississippi State, MS. (Poster, I Place).
 28. Poudel, S., Sankarapillai, L. V., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Morpho-physiological and yield responses of cowpea to drought stress., Spring Graduate Research Symposium, Mississippi State, MS. (Poster II Place).
 29. **Adhikari B.**, Sankarapillai, L. V., Dhillon J.S., Reddy, K.N., Reddy, K.R., & Bheemanahalli, R. (2024). Soybean response to early- and late-season waterlogging. Southern Branch of American Society of Agronomy, Atlanta, GA. (Oral presentation).

30. Bista, M. K., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Drought resilience of cotton breeding lines during reproductive stage. Southern Branch of American Society of Agronomy, Atlanta, GA. (Oral presentation).
31. Poudel, S., Sankarapillai, L. V., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Morphological, physiological and yield responses of cowpea to drought stress. Southern Branch of American Society of Agronomy, Atlanta, GA. (Oral presentation).
32. Sankarapillai, L. V., **Adhikari B.**, Stetina, S. R., Reddy, K. R., & Bheemanahalli, R. (2024). Physiological and yield responses of soybean cultivars to elevated night temperature during reproductive stage. Southern Branch of American Society of Agronomy, Atlanta, GA. (Oral presentation).
33. Bista, M. K., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Effects of heat stress on cotton physiology and yield. Southern Branch of American Society of Agronomy, Atlanta, GA. (Poster presentation)
34. Poudel, S., Sankarapillai, L. V., **Adhikari B.**, Reddy, K. R., & Bheemanahalli, R. (2024) Morpho-physiological and yield responses of cowpea to drought stress., Southern Branch of American Society of Agronomy, Atlanta, GA. (Poster presentation).
35. **Adhikari, B.**, Sankarapillai, L. V., Vennam, R.D., Dhillon, J.S., Reddy, K.N., Reddy, K.R., & Bheemanahalli, R. 2023. Resilience of Row Crops to Short-Term Waterlogging during Early Growth and Development. ASA, CSSA, SSSA International Annual Meeting (Poster presentation)
36. **Adhikari, B.**, Wilson, J., Bheemanahalli, R. 2023. Morpho-Physiological and Biochemical Responses of Hydroponically Grown Romaine Lettuce to Salt Stress. ASA, CSSA, SSSA International Annual Meeting (Oral presentation)
37. Bista, M.K., **Adhikari, B.**, Sankarapillai, L. V., Reddy, K.R., Jenkins, J., & Bheemanahalli, R. 2023. Quantifying the Impact of Drought Stress on Physiology and Yield of Cotton. ASA, CSSA, SSSA International Annual Meeting (Poster presentation)
38. Poudel, S., Sankarapillai, L. V., **Adhikari, B.**, Bista, M.K., Reddy, K.R., & Bheemanahalli, R. 2023. Response of Cowpea to Drought Stress during Early Vegetative Growth and Development. ASA, CSSA, SSSA International Annual Meeting (Poster presentation)
39. **Adhikari, B.**, J. Wilson, and R. Bheemanahalli, 2023. Decoding genotypic variability in romaine lettuce for improved salt stress resilience. Mississippi Academy Sciences, Summer Science and Engineering Symposium (Poster presentation)
40. Hilaire, P., **Adhikari, B.**, R.V. Reddy, R.V., Reddy, K.R., & Bheemanahalli, R. 2023. Impact of salt stress on early-season growth and development in sorghum. Mississippi Academy Sciences, Summer Science and Engineering Symposium (Poster presentation)
41. **Adhikari, B.**, Barickman, T.C., & Bheemanahalli, R. 2023. Response of lettuce (*Lactuca sativa* L.) to salt stress. Mississippi State University Spring Graduate Symposium (Oral presentation)
42. **Adhikari, B.**, Barickman, T.C., & Bheemanahalli, R. 2023. Seed Priming attenuates the salt stress impact and improves yield in lettuce. Mississippi State University Spring Graduate Symposium (Poster presentation)
43. **Adhikari, B.**, Barickman, T.C., & Bheemanahalli, R. 2023. Response of lettuce (*Lactuca sativa* L.) to salt stress. Mississippi Academy Sciences Conference (Oral presentation)
44. Olorunwa, O.J., Brazel, S.R., **Adhikari, B.**, Popescu S. C., & Popescu, G., & Barickman, T.C. 2022. Short-Term Waterlogging Alters Photosynthesis, Chlorophyll Fluorescence, and Response to CO₂ and Light Intensity in Cucumber (*Cucumis sativus* L.) Seedlings. American Society of Horticultural Sciences Annual Conference (Oral presentation)
45. **Adhikari, B.**, Olorunwa, O.J., Horgan, T., & Barickman, T.C. 2022. Seed Priming and Salt Stress Impacts Germination, Phenotypic Traits, and Biomass Accumulation of Lettuce (*Lactuca sativa* L.), Southern Region American Society of Horticultural Sciences Annual Conference (Poster presentation)
46. **Adhikari, B.**, Olorunwa, O.J., Horgan, T., & Barickman, T.C. 2022. Photosynthetic Efficiency and Chlorophyll Fluorescence are Impacted by Seed Priming and Salt Stress in Lettuce (*Lactuca sativa*

- L.). Southern Region American Society of Horticultural Sciences Annual Conference (Poster Competition)
47. Blanche, T., **Adhikari, B.**, Olorunwa, O.J., Horgan, T., & Barickman, T.C. 2022. Elevated Levels of Iron influence physiological Performance, Yield, and Soluble Sugar and Phytonutrient Concentrations of Lettuce (*Lactuca sativa* L.). Southern Region American Society of Horticultural Sciences Annual Conference (Poster Competition)
 48. **Adhikari, B.**, Olorunwa, O.J., & Barickman, T.C. 2022. Salt Stress Negatively Affects the Growth and Development, Net Photosynthetic, Chlorophyll Fluorescence, and Biochemical Responses in Lettuce (*Lactuca sativa* L.). American Society of Horticultural Sciences Annual Conference (Poster Competition)
 49. **Adhikari, B.**, Olorunwa, O.J., & Barickman, T.C. 2022. Salt Stress Induces Morpho-Physiological and Biochemical Responses in Lettuce (*Lactuca sativa* L.). American Society of Horticultural Sciences Annual Conference (Poster Competition)
 50. Olorunwa, O.J., **Adhikari, B.**, Shi, A., & Barickman, T.C. 2022. "Screening of cowpea (*Vigna unguiculata* L.) genotypes for waterlogging tolerance using morpho-physiological traits at early growth stage" Southern Region American Society of Horticultural Sciences Annual Conference (Oral Competition)
 51. Olorunwa, O.J., **Adhikari, B.**, & Barickman, T.C. 2022. "Evaluating Waterlogging Stress on the Morphological and Gas Exchange Parameters of Cowpea (*Vigna unguiculata* (L.) Walp.) Genotypes Under Field Conditions" Southern Region American Society of Horticultural Sciences Annual Conference (Poster Competition)
 52. **Adhikari, B.**, & Barickman, T.C. 2021. Morphological and Physiological Response of Different Lettuce Cultivars Under Salt Stress (Poster competition)
 53. Vashisth, T. & **Adhikari, B.**, 2021. Use of Plant Growth Regulators to Onset and Release Dormancy in Low-Chill Peach Cultivars. American Society of Horticultural Sciences (Oral Presentation)
 54. **Adhikari, B.**, Dutt, M., & Vashisth, T. 2020. Phytochemical Profile in Finger Lime (*Citrus australasica*), Southern Region American Society of Horticultural Sciences Annual Conference (Poster presentation)
 55. **Adhikari, B.**, Williamson, J.G., Sarkhosh, A., and Vashisth, T. 2020. Use of Plant Growth Regulators to Induce Uniform Flowering in Low-chill Peach Cultivar 'UFBest', Southern Region American Society of Horticultural Science (Oral presentation)
 56. **Adhikari, B.**, & Vashisth, T. 2020. Evaluation of the effect of Absciscic acid and Hydrogen cyanamide in inducing uniform flowering in low-chill peach cultivar 'UFBest', Southern Region American Society of Horticultural Sciences Annual Conference (Oral presentation)
 57. **Adhikari, B.** & Vashisth, T. 2019. Evaluation of the effect of different rates of Hydrogen cyanamide in low-chill peach cultivars, American Society of Horticultural Sciences Annual Conference, (Ignite competition)
 58. **Adhikari, B.** & Vashisth, T. 2019. Evaluation of plant growth regulators to induce uniform flowering in low-chill peach cultivar 'UFBest', American Society of Horticultural Sciences Annual Conference (Poster Presentation)

Reviewed paper journals: 24 as a peer reviewer for journals and three as an independent reviewer for USDA_ARS

1. Nature: Scientific Reports
2. Frontiers in Plant Science
3. The Crop Journal
4. Plant Stress
5. Physiologia Plantarum
6. Journal of Agriculture and Food Research
7. Biologia Plantarum
8. Agrosystems, Geosciences & Environment

